

Laboratory Report

Course ID: CPS 3498

Lab 1: Identity and security

Student: Jason Houston

Instructor: Dr. Bin Hu

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Description

This section describes the objective of the lab based on your understanding from the actual laboratory work. You should provide your **own statements**, but not copy the information listed in the "Description" section of the Lab Procedure document.

In addition, you should provide in the section the **system configuration information** including at least: Computer and its operating system version number, location (building on campus or township if off-campus) where the lab performed, and type of network connection and Internet Service Provider (ISP) used to connect to the Internet.

If any of these conditions are changed while conducting the lab procedure, you should note immediately the change in the **"Procedure and Notes"** section next to the procedure that is performed.

Procedure and Notes

1. Outcomes in a Laboratory Report should be reproducible based on the same settings. Hence, this section is an itemized list of actions and notes you took by following the lab procedure.

   List of Actions Performed

   Lab 1.1 Windows Client IP Configuration

1. Logged into a Windows 11 computer with administrative privileges.
2. Opened Command Prompt by searching for **cmd** in the Windows search bar.
3. Executed the following command:

   BAT

   ipconfig /?
4. Reviewed the available ipconfig command options and identified common administrative options such as:
   - /all
   - /release
   - /renew

- /flushdns

5. Executed:

   BAT

   ipconfig

6. Recorded the following network configuration information:
   - IPv4 Address
   - Subnet Mask
   - Default Gateway
7. Reviewed the purpose of the subnet mask and noted that it identifies the network portion and host portion of an IP address.
8. Executed:

   BAT

   ipconfig /all

9. Recorded additional network information:
   - Physical Address (MAC Address)
   - DNS Server Address
   - DHCP Status
   - Network Adapter Information
10. Opened the network adapter properties by navigating to:
    - Control Panel
    - Network and Internet
    - Network and Sharing Center
    - Change Adapter Settings
    - Network Adapter Properties
    - Internet Protocol Version 4 (TCP/IPv4)

11. Compared the IP address shown in the adapter settings with the IP address reported by the ipconfig command.
12. Verified that the computer was configured to obtain an IP address automatically (DHCP).
13. Reviewed the procedure for assigning a static IP address but did not permanently change the network configuration to avoid disrupting connectivity on the personal computer.
14. Documented the expected result of assigning a static IP address, which would temporarily change the IPv4 configuration until restored.

Lab 1.2 IP Network Configuration

Part A - Ping and Tracert

15. Executed:

    BAT

    ping /?

16. Reviewed the available Ping command options.
17. Determined the Default Gateway address using:

    BAT

    ipconfig

18. Executed a ping test to the Default Gateway:

    BAT

    ping [Default Gateway Address]

19. Recorded:

- Round-trip response times
- TTL (Time To Live) values
- Packet statistics

20. Verified that responses originated from the correct destination by matching the responding IP address with the target IP address.
21. Executed a traceroute to the Kean University website:

BAT

tracert www.kean.edu

22. Recorded:

- Number of hops
- Intermediate routers
- Response times

23. Repeated the traceroute multiple times and observed route variations.
24. Created a simple network path diagram based on the traceroute results.
25. Executed a traceroute to Google:

BAT

tracert www.google.com

26. Recorded:

- Number of hops
- Intermediate nodes
- Longest response time observed

27. Compared the Google route with the Kean University route and documented differences.

Lab 1.2 IP Network Configuration

Part B - ARP

28. Executed:

BAT

arp /?

29. Reviewed available ARP command options.
30. Identified that the following command displays the current ARP cache:

BAT

arp -a

31. Executed:

    BAT

    arp -a

32. Recorded:

- IP addresses
- Corresponding MAC addresses
- Dynamic ARP entries

33. Reviewed the procedure for deleting ARP cache entries using:

    BAT

    arp -d

34. Chose not to modify the ARP cache on the personal device to avoid unnecessary network interruptions.

35. Documented that deleting the ARP cache would remove stored IP-to-MAC mappings and that Windows would automatically rebuild the table when network communication resumes.

36. Executed:

    BAT

    arp -a

    again to verify current ARP entries.

Conclusion

41. Successfully examined the computer's IP configuration, network adapter settings, DNS information, MAC address information, connectivity status, routing paths, and ARP cache information without making permanent changes to the system configuration.

2. It should provide any data/information that you are asked to note or observe at that particular procedure. Some screen shots may be attached as a proof of your note and provide your instructor to determine the root cause, if your note would be incorrect.

3. If the laboratory requires to develop a program code, you should list your program code in this section.

4. You can list any problems you encountered during the lab. Discuss why it occurred and how to get it right for the correct response, if you did.

   I did not encounter any issues

Results and Reports

1. This section is for you to discuss your final results and/or report from your laboratory experience and conclusion.

2. If there are any files prepared as the results of the lab, make sure they are accessible electronically.

```
C:\Users\dstdi>tracert www.google.com

Tracing route to www.google.com [142.251.151.119]
over a maximum of 30 hops:

  1    15 ms     2 ms     1 ms  openrg.home [192.168.1.1]
  2    10 ms    16 ms    12 ms  10.240.170.53
  3    35 ms    25 ms    12 ms  ubr103-ge3-0-1.cmts.nwrknj.cv.net [67
  4    13 ms    13 ms    11 ms  ool-4353fa80.dyn.optonline.net [67.83
  5    27 ms    13 ms    13 ms  67.59.239.119
  6    28 ms    14 ms    17 ms  64.15.6.4
  7    20 ms    11 ms    15 ms  142.251.200.104
  8    27 ms    12 ms    14 ms  142.251.151.119

Trace complete.
```

```
C:\Users\dstdi>tracert www.kean.edu

Tracing route to web.kean.edu [131.125.16.38]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  openrg.home [192.168.1.1]
  2    22 ms    10 ms    12 ms  10.240.170.53
  3    24 ms    13 ms    11 ms  67.59.253.136
  4    25 ms    12 ms    13 ms  dstswr1-ge3-16.rh.wlwknj.cv.net [67.8
  5    16 ms    24 ms    13 ms  rtr1-tg10-2.mhe.prnynj.cv.net [64.15.
  6    31 ms    14 ms    24 ms  64.15.6.8
  7     *        *        *     Request timed out.
  8    16 ms    14 ms    15 ms  ae3.cr9-nyc2.ip4.gtt.net [208.116.158
  9    28 ms    13 ms    15 ms  ae0.cr3-nyc12.ip4.gtt.net [89.149.142
 10    28 ms    14 ms    16 ms  ip4.gtt.net [209.120.154.146]
 11     *        *        *     Request timed out.
 12     *        *        *     Request timed out.
 13     *        *        *     Request timed out.
 14     *        *        *     Request timed out.
 15     *        *        *     Request timed out.
 16     *        *        *     Request timed out.
 17     *        *        *     Request timed out.
 18     *        *        *     Request timed out.
 19     *        *        *     Request timed out.
 20     *        *        *     Request timed out.
 21     *        *        *     Request timed out.
 22     *        *        *     Request timed out.
 23     *        *        *     Request timed out.
 24     *        *        *     Request timed out.
 25     *        *        *     Request timed out.
 26     *        *        *     Request timed out.
 27     *        *        *     Request timed out.
 28     *        *        *     Request timed out.
 29     *        *        *     Request timed out.
 30     *        *        *     Request timed out.

Trace complete.
```

```
C:\Users\dstdi>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64
Reply from 192.168.1.1: bytes=32 time=3ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Users\dstdi>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\Users\dstdi>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : LAPTOP-Q8KU3PPL
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet A
   Physical Address. . . . . . . . . : 0A-00-27-00-00-08
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::e4d8:723b:d1bf:94da%8(Pr
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 671744039
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2E-94-E2-4B-F8-54-
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual
   Physical Address. . . . . . . . . : FA-54-F6-23-C1-78
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual
   Physical Address. . . . . . . . . : FE-54-F6-23-C1-78
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : home
```

```
Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

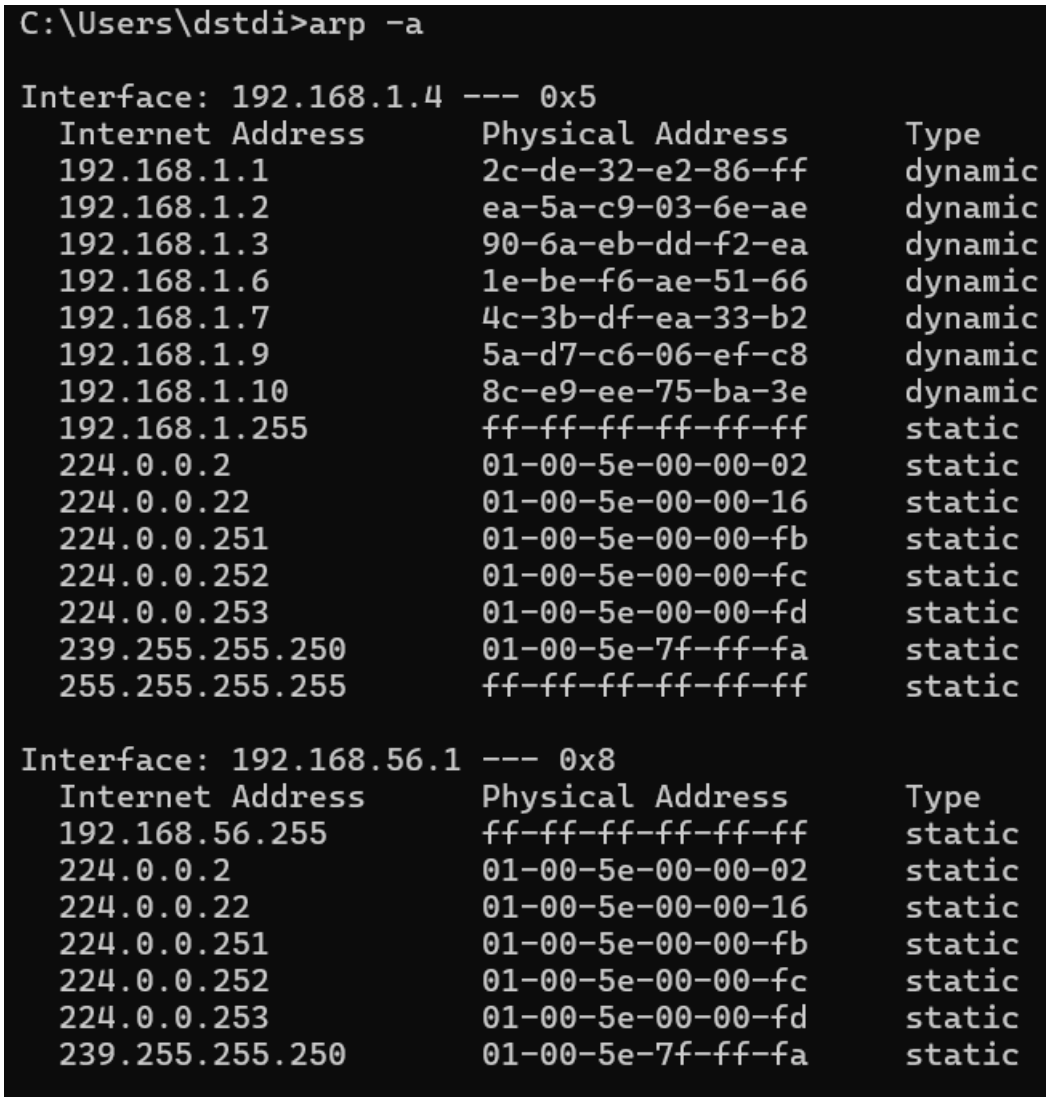
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : home
   IPv6 Address. . . . . . . . . . . : fd22:4e39:e630:1:ae7f:28f9:781
   Temporary IPv6 Address. . . . . . : fd22:4e39:e630:1:8923:7a34:1a5
   Temporary IPv6 Address. . . . . . : fd22:4e39:e630:1:b99a:6789:f41
   Temporary IPv6 Address. . . . . . : fd22:4e39:e630:1:d42c:2566:496
   Temporary IPv6 Address. . . . . . : fd22:4e39:e630:1:f10f:e446:88b
   Link-local IPv6 Address . . . . . : fe80::5d20:af94:8779:1efd%5
   IPv4 Address. . . . . . . . . . . : 192.168.1.4
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```



```
C:\Users\dstdi>arp -a

Interface: 192.168.1.4 --- 0x5
  Internet Address      Physical Address      Type
  192.168.1.1           2c-de-32-e2-86-ff     dynamic
  192.168.1.2           ea-5a-c9-03-6e-ae     dynamic
  192.168.1.3           90-6a-eb-dd-f2-ea     dynamic
  192.168.1.6           1e-be-f6-ae-51-66     dynamic
  192.168.1.7           4c-3b-df-ea-33-b2     dynamic
  192.168.1.9           5a-d7-c6-06-ef-c8     dynamic
  192.168.1.10          8c-e9-ee-75-ba-3e     dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.0.0.253           01-00-5e-00-00-fd     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0x8
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.0.0.253           01-00-5e-00-00-fd     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```

Lab 1.1: Windows Client IP Configuration (ipconfig)

- **Which options do you think would be most useful for an administrator?**
    - /all (displays full configuration details including physical MAC addresses, DNS servers, and DHCP state) and /renew or /release (to manage dynamic IP lease allocations) are the most useful.
- **Which option would you use to obtain an IP configuration from a Dynamic Host Configuration Protocol (DHCP) server?**
    - ipconfig /renew
- **What is your IP address?**
    - 192.168.1.4 (Wi-Fi adapter)

- **What is your subnet mask? What is the purpose of an IP address mask?**
    - Subnet Mask: 255.255.255.0
    - **Purpose:** The subnet mask identifies which portion of the IP address represents the network ID and which portion represents the host ID, allowing devices to determine whether communication is local (on the same LAN) or requires a router/gateway.
- **What is the MAC address of your computer?**
    - For the VirtualBox Host-Only Ethernet adapter: 0A-00-27-00-00-08. (For the Wi-Fi adapter, the physical address can be referenced via adapter details if needed).
- **What is your DNS server address?**
    - 192.168.1.1

Lab 1.2: IP Network Configuration

Lab A: ping and tracert

- **What is the time value observed for all four replies? (pinging gateway 192.168.1.1)**
    - Round trip times were 1ms, 2ms, 2ms, and 3ms (average of 2ms).
- **What is the TTL (Time To Live) value? What does this number refer to?**
    - TTL Value: 64
    - **Reference:** It defines the maximum number of router hops a packet can traverse before being discarded, preventing infinite routing loops.
- **How can you be sure that this response is actually coming from the correct computer?**
    - By pinging the exact local gateway IP (192.168.1.1) or verifying the matching MAC address resolved through the ARP cache table.
- **tracert www.kean.edu observations:**
    - **How many actual intermediate hops?** Around 6 responding active hops before reaching external transit nodes and timeouts (web.kean.edu resolves to 131.125.16.38).
    - **Which intermediate node takes the longest time?** Hop 2 (10.240.170.53) and initial ISP transit edges fluctuated up to 22ms–35ms.

- **tracert www.google.com observations:**
    - **How many actual intermediate hops?** 8 hops to reach Google's destination server (142.251.151.119).
    - **Which intermediate node takes the longest time?** Hop 3 (ubr103-ge3-0-1.cmts.nwrknj.cv.net) showed initial spikes around 35ms.

Lab B: ARP (Address Resolution Protocol)

- **At the command prompt type arp /? and press ENTER. Which option displays the current ARP entries?**
    - The -a option (arp -a) displays the current ARP cache entries.
- **Observations from arp -a:**
    - Successfully populated dynamic entries mapping local IP addresses (such as gateway 192.168.1.1) to their respective physical MAC addresses (e.g., 2c-de-32-e2-86-ff).

Reference and Acknowledgement

With or without encountering any difficulty, if you surfed the Internet to find some hints or to learn new information and/or technology to help you conducting the lab, you should include the information sources, such as the URLs, citations, or books.

I did not.

If you received assistance from anyone (e.g. working together with some friend for the lab), you should acknowledge people’s contribution.

No assistance was used.